Data Analysis: Assessing The Role Of Frontex's Aerial Operations In Enabling Pushbacks In The Central Mediterranean

Stanislas Michel, Geospatial Researcher
November 2026

LIMINAL
INVESTIGATING IM–MOBILITIES AND BORDER VIOLENCE

liminal-lab.org
@liminal_laboratory

European Research Council
Established by the European Commission

Funded by the European Union (ERC Starting Grant, Project 101042338 —HEMIG)

Summary 3

Context 4

Data 5

Methodology and Results 7

Discussion of Results 15

Conclusion 19

European Research Council
Established by the European Commission

Funded by the European Union (ERC Starting Grant, Project 101042338 —HEMIG)

Summary

Due to the lack of safe and legal pathways to reach Europe, many migrants are forced to risk their lives daily by attempting to cross the Central Mediterranean from Libya. While it is widely recognized that the Libyan Coast Guard (LYCG) violently intercepts these boats and returns people to Libya against their will, the role of Frontex — the European Border and Coast Guard Agency — in facilitating these operations remains less clear. Available evidence indicates that Frontex uses aerial surveillance in the region to relay information to the LYCG, effectively enabling violent interceptions. This occurs despite knowledge that those intercepted are likely to be sent back to Libya, where they face arbitrary imprisonment, abuse, and exploitation.

This study aims at demonstrating, through data analysis, the extent of Frontex's complicity and responsibility in these forced interceptions and returns, which we will refer to as "pushbacks".[1] To do so, multiple data sources were gathered and analysed: UNHCR monthly records of interceptions and disembarkations in Libya (2016-2023); IOM Libya's daily dataset on interceptions (2020–2022); and successive datasets obtained from Frontex's Joint Operation Reporting Application (JORA) [IC1.1]database (2016–2023), which were harmonised and filtered to identify interceptions and arrivals.

Together these sources allow us to assess the degree of Frontex's operational involvement in pushbacks, the completeness of the Agency's own reporting and the extent to which migrants attempting to cross the Central Mediterranean are "pushed back" to Libya.

All in all, this study demonstrates that Frontex's aerial assets have consistently enabled Libyan Coast Guard interceptions while playing only a marginal role in facilitating operations that resulted in the effective rescue and disembarkation of passengers to a place of safety (in this case, Italy and Malta) as mandated by international law. Independent evidence from non-governmental organisations also shows that Frontex does not record all incidents in which its aircraft are operationally present, further undermining accountability. Finally, the study accounts for Frontex's increasing lack of transparency, which over time has made it progressively impossible to conduct a meaningful analysis of its own data obtained through Freedom of Information procedures, thus preventing independent scrutiny of its aerial operations.

1 The terminology used to describe these practices is varied and contested across the fields of international law and politics. For the purpose of this study, "interception" refers specifically to the act of a state stopping a vessel at sea, while "forced return" is used to indicate the forced removal of an individual to their country of departure. We will refer to these operations together as "pushbacks" (for a definition of which, see here: https://home-affairs.ec.europa.eu/networks/european-migration-network-emn/emn-asylum-and-migration-glossary/glossary/push-back_en).

Context

At the request of LIMINAL, data analysis was conducted to examine the role of Frontex's aerial assets in the interception of migrants by Libyan authorities in the Central Mediterranean. The analysis mainly focused on the period from 2016, i.e. shortly before the Agency started to conduct systematic aerial surveillance operations in the Central Mediterranean, to June 2025. Migrants and asylum seekers frequently attempt to cross the Central Mediterranean Sea from Libya to reach safety in Europe. These crossings are often made in unseaworthy vessels, making the journey extremely dangerous. However, European cooperation with Libyan authorities, including through the support and involvement of Frontex (the European Border and Coast Guard Agency), has significantly affected these migration routes.

Reports and documentation indicate that Frontex operates aerial surveillance assets, including several aircraft and a drone, to monitor the Central Mediterranean. These assets are believed to relay information about migrant boats to the Libyan Coast Guard (LYCG), who then intercept and return migrants to Libya – a country where many face arbitrary detention, abuse, and other human rights violations.[2]

The core objective of this statistical analysis is to determine whether a correlation exists between Frontex aerial surveillance activities, and the number of interceptions carried out by the LYCG. Specifically, the analysis seeks to investigate how closely Frontex's surveillance operations align with interception events, i.e pullbacks to Libya and Tunisia, as compared to migrant's safe arrivals in Italy and Malta.

Preliminary results suggest a strong correlation between the deployment of Frontex aerial assets and subsequent interceptions by the LYCG. In numerous instances, aircraft activity in specific maritime zones closely preceded interception events, implying potential operational coordination or information sharing. In parallel, Frontex's involvement in events resulting in arrivals to Italy seems drastically smaller. At the same time, Frontex's systematic censorship of its own datasets, through frequent redaction of key elements, has made it increasingly harder to assess the agency's role in these operations.

2 Judith Sunderland and Lorenzo Pezzani, "Airborne Complicity," Human Rights Watch, December 8, 2022, https://www.hrw.org/video-photos/interactive/2022/12/08/airborne-complicity-frontex-aerial-surveillance-enables-abuse.

Data

In order to achieve the objectives outlined above, we compiled and cross-referenced multiple data sources. Existing relevant datasets, i.e. those containing information about migrant crossings, interceptions, forced returns and arrivals in the Central Mediterranean, are not always consistent, nor complete. The following subsection describes these datasets, exposing their limitations, and explaining the criteria that guided our decisions in making them comparable and legible.

1. UNHCR and IOM data

Both of these well-known international organizations regularly collect data about interceptions and returns to Libya, which are made publicly available via their websites. They maintain a significant presence at Libyan disembarkation points and have direct contact with Libyan authorities, thus offering the most reliable and comprehensive datasets in a context where access of reliable information is scarce and difficult. Yet their data collection processes and methodologies are different.

UNHCR provides monthly counts of people intercepted by the Libyan Coast Guard (LYCG) and disembarked in Libya for 2016–2025 but does not report the number of boats involved. IOM Libya instead offers daily counts of intercepted migrants for 2020–2022, with partial information on boats, but its boat figures do not always match other interception datasets[IC8.1]. In this study we use UNHCR as our primary source because it provides the most consistent time series and the longest coverage for the area where people are actually disembarked. The differences between UNHCR and IOM data are modest (typically only a few percent), so this choice does not materially change levels or trends but helps to maintain consistency over the period of study.

All considered, UNHCR and IOM data provide a solid and clear baseline against which the accuracy of the more opaque Frontex JORA dataset, which we discuss below, could be tested.

2. Frontex JORA (Joint Operations Reporting Application) database

JORA is a standardized information system established and maintained by Frontex in collaboration with EU Member States and other EU institutions. In this database, European officials taking part in Frontex Joint Operations archive and share information about border crossing “incidents” and other significant events taking place across all European borders. Since the surveillance of the EU’s borders is one of the Agency’s key objectives and because the data is collected by the agency itself for internal use, this dataset supposedly contains one of the most complete records of its aerial surveillance activities in the Central Mediterranean. The data from JORA, however, is not readily legible. Therefore, as explained in more detail below, a significant part of the work of this study has involved conducting a thorough analysis and interpretation of the JORA dataset.

Data used in LIMINAL’s analysis of JORA was obtained through multiple Freedom of Information (FOI) requests. The first, filed by investigative journalist Emmanuel Freudenthal, covered the period from mid-September 2016 to early September 2021.[3] LIMINAL later submitted a follow-up request[4] to extend the dataset to January 2024,[5] when our FOI was lodged. Despite both requests

3 Arne Semsrott, *incidents_20160914* in Frontex Public Register of Documents, distributed by OCCRP Aleph Pro, https://aleph.occrp.org/entities/52913ea4c5b-70cba04cd39208e19c42adfb8e-2ca.9928031 9d57d7ce5675614290d06b3242df77666.

4 https://www.asktheeu.org/request/jora_variables_2021_2024#incoming-52165

5 European Border and Coast Guard Agency. JORA variables 2021-2024. FOI Requested by Giovanna Reder. AsktheEU, 2024. https://www.asktheeu.org/request/jora_varibles_2021_2024#incoming-52165.

seeking identical information from the same Frontex repository, the datasets differed significantly, not only in structure but also in content. Additionally, Frontex's response included data only up to January 2023—much of it heavily redacted.

A subsequent FOI request[6] aimed to obtain the missing information, but Frontex rejected it, citing public security risks, ongoing evaluations, and the sensitivity of operational material. LIMINAL filed another request in April 2025, which was also denied. We then submitted a confirmatory application,[7] arguing that since the operational period had ended, the data should be disclosed. While Frontex eventually released additional material, the documents were so extensively redacted that they were unusable for our analysis.

An increasing amount of critical information, which was present in the first dataset (hereafter, JORA 1), had been redacted upon the release of the[IC10.1] second dataset (hereafter, JORA 2). In particular, columns that had been present in JORA 1 were redacted in JORA 2, stripping the data of crucial variables; several other columns were renamed; and the typologies or categorization of values within columns were altered. These inconsistencies introduced a layer of complexity in conducting a continuous and coherent analysis across the full temporal scope of the study and decreased the level of confidence with which we were able to link interception events to Frontex aerial surveillance activities. These issues were substantially exacerbated in the third release (hereafter, JORA 3). The withholding of critical fields concerning key operational variables such as disembarkation place, detection initiation, Frontex asset involvement and date of incident, has in fact prevented the possibility to assess Frontex's involvement in pushbacks. This is despite the fact that JORA 3 contained at least 6,100 operational records concerning Joint Operation Themis, i.e. the operation that Frontex conducts in the Central Mediterranean, including incidents involving 240,000 people and 240 deaths and disappearances. Crucially, the fact that the data withheld in the second and third dataset was initially supplied suggests that this denial of access can not be justified by security reasons but rather serves to obstruct independent public oversight of the activities of a key EU agency.

In light of all the issues and discrepancies listed above, and as described in more detail below, we applied an in-depth curation and standardisation process with the aim of producing a coherent and temporally consistent dataset over the course of the almost seven-year study period, so as not to introduce methodological bias resulting from the internal inconsistencies between the datasets.

JORA 1	14.09.2016 - 06.09.2021
JORA 2	01.01.2021 - 07.10.2023
JORA 3	01.01.2023 - 18.07.2025

6 European Border and Coast Guard Agency. JORA variables 2021-2024 (II). FOI Requested by Giovanna Reder. AsktheEU, 2024.https://www.asktheeu.org/request/jora_variables_2021_2024_ii#incoming-54298.

7 European Border and Coast Guard Agency. JORA Variables 2023-2025. FOI Requested by Giovanna Reder. AsktheEU, 2025. https://www.asktheeu.org/request/jora_variables_2023_2025#incoming-59927.

Methodology And Results

Our analysis focused specifically on two types of events: “pushbacks” (the interception of migrant boats at sea and the forced return of their passengers to Libya and Tunisia) and “arrivals” (the landing of migrant boats departing from Libya and Tunisia to Italy).

The purpose of the study was twofold:

a) Establish the number of relevant events (“pushbacks” and “arrivals”, as defined above) recorded in Frontex own database

b) Assess the role played by Frontex aerial surveillance operations in each of these events (i.e. whether Frontex was merely aware of the event, contributed to it, or initiated it) and the degree of confidence in our attribution of responsibility[LP11.1], based on the quality of the available evidence.

In order to reach aim a) we applied a series of successive filters, so as to isolate a subset of relevant events despite inconsistencies in the dataset structure and in reporting practices. This was necessary because, as already recalled, the JORA database includes information about many different types of events that have occurred across all of the EU’s external borders. Objective b) was achieved through two different grading systems, one concerning pushbacks and the other arrivals. In both cases, we categorised events on a scale measuring from events of which Frontex was simply aware of (but in which the Agency might not have played a direct role) to events in which Frontex aerial detection triggered the event (and for which therefore the Agency bears a higher degree of responsibility). The final goal was to provide an assessment of the degree of involvement — and therefore responsibility for —each of the events listed.

We now turn to a more in-depth description of each step of our analysis in relation to each of the two categories of events considered.

1. PUSHBACKS

a) Establish the number of pushbacks to Libya and Tunisia recorded in the Frontex JORA database

In order to fulfil aim a), we undertook the following operations. To begin with, a consistent temporal marker was established for each incident. Given irregularities and missing values across the dataset, the date field was derived by prioritising the available timestamps in the following order: DetectionDate, followed by InterceptionDate, and finally ReportingDate where the former were not available. This ensured that each record was anchored in time for the purpose of longitudinal analysis.

The number of events corresponds to the number of boats that arrived in Italy or were pulled back, while the total number of people who arrived in Italy or were intercepted was calculated by extracting the maximum value across several related columns—num_total_irreg_migrants, num_total_persons, num_EstimatedPOB, and num_I_MigrantsDeterred—to capture the most complete count available per incident. From this maximum value, any reported death cases (num_I_DeathCases) were subtracted to estimate the number of individuals who were intercepted and not deceased – so as to remain consistent with UNHCR data.

We then proceeded to apply a series of successive filters on the overall JORA database to identify relevant events, each filter being applied to the output of the previous filtered events[IC12.1]. Filters had to be adapted to the first and second [IC13.1]JORA dataset considered in the study due to their internal differences, as detailed below.

1 Filter 1 > Countries of departure = Libya and Tunisia

First, Filter 1 restricted the dataset to incidents involving departures from Libya, Tunisia, or where the departure country was unknown or not recorded. Specifically, values of CountryofDeparture were limited to: 'Libya', 'Tunisia', 'Unknown', 'Not applicable', 'Not Identified by screener', or left empty (i.e., missing data).

2 Filter 2 > Countries of disembarkation = Libya and Tunisia

For JORA 1, covering the period September 2016 to September 2021, Filter 2 focused on disembarkation locations, retaining only those events where migrants were brought ashore in Libya or Tunisia. Due to inconsistent spelling and data entry variations, a comprehensive list of disembarkation locations was compiled. This included multiple permutations and transliterations of known coastal cities and regions such as Tripoli, Al Khums, Zawiya, Sfax, Zarzis, Misurata, and others. The intention was to capture all records that could plausibly reflect disembarkation events in Libya or Tunisia, even where place names were inconsistently formatted.

This second filter, however, could not be applied to JORA 2 (September 2021 to January 2023). This is because information concerning the disembarkation location was not made available in the second dataset. To make up for this absence, and the subsequent limitation imposed on our analysis, Filter 3 was applied to identify incidents that Frontex categorised as interceptions or preventions of departure, which are typically associated with attempts to stop unseaworthy boats from leaving the territorial waters of Libya or Tunisia

3 Filter 3 > Type of Incident

For JORA 1, Filter 3 was designed to capture a specific subset of cases potentially classified differently within the dataset. It identified incidents with a PrimaryIncidentType of 'Prevention of departure' or 'Third Country Action'.

For JORA 2, incidents were selected if:

- The interception place was labelled as “Prevention of departure”, or
- The primary incident type was recorded as “Prevention of departure” or “Third Country Action”.

These criteria were selected to ensure the inclusion of all records that Frontex considered relevant to maritime interception activities, even if no direct reference was made to the final place of disembarkation. These labels were chosen on the basis of a thorough analysis of the JORA 1 dataset, in which interception incidents take these values for these columns.[IC14.1]
A potential caveat in this dataset stems from the lack of disembarkation data and the presence of entries that include a “Country of Destination” field listing countries such as Italy, France, Germany, Belgium, the Netherlands, the United Kingdom, Spain, Switzerland, or Unknown. While these entries suggest an intended destination in Europe, their presence does not definitively confirm where the boats ultimately arrived or whether they were intercepted before reaching European shores.

In light of the process described above, we can confidently say that all entries retained through this filtering process describe departures from Libya or Tunisia and involve migrant boats that were either intercepted or prevented from departing—falling squarely within the scope of this study. As such, the filtering system is appropriate for quantifying the number of maritime interceptions documented by Frontex, whether the agency was directly operationally involved.

Results

This harmonised filtering approach, applied consistently across both datasets (pre- and post-September 2021), enables to extract a coherent subset of relevant events over the entire period covered by the JORA 1 and JORA 2 datasets (i.e. 2016–2023).

JORA extract	Filtered incidents	People recorded as intercepted
JORA 1	748	50,060
JORA 2	362	19,749
Total (combined)	**1,110**	**69,809**

This analysis also provides a basis for examining Frontex's awareness and documentation of interceptions, independently of its degree of involvement or the final outcomes of the incidents.

b) Assess the role played by Frontex aerial surveillance operations in these pushbacks and the degree of confidence in our attribution of responsibility

To refine the analysis of Frontex's involvement in maritime pushback incidents in the Central Mediterranean, a graded system of operational attribution was developed. This system allows for a structured categorisation of incidents according to the quality of the evidence available [LP15.1]linking Frontex aerial surveillance to each interception and Frontex's degree of involvement in each of these.

The baseline level of documentation, Grade 4, includes every pushback incident recorded in Frontex's JORA database after filtering for relevant geographic and operational parameters (as described in previous sections). From this baseline, increasingly stringent criteria were applied to establish more direct links between specific incidents and Frontex surveillance operations, culminating in Grade 1, which indicates the highest level of involvement in a pushback, assessed with the strongest available evidence.

Grade 4 – Frontex knowledge of a pushback but no specific evidence of Frontex involvement

Grade 4 includes all maritime puhbacks accounted for in Frontex's own JORA database, based on the final dataset resulting from the geographic and typological filtering processes. This category reflects events that Frontex acknowledges in its reporting system but does not necessarily indicate direct involvement in such events through either surveillance or operational support.

Grade 3 – Plausible / Frontex detected the boat at some point and could have initiated the pushback

To isolate cases in which Frontex played a more active role, a fourth filter was introduced to the Grade 4 dataset: only incidents where the field DetectionByFrxAsset is marked as "Yes" were retained. This indicates that Frontex's assets—such as aircraft or drones—were directly involved in detecting the vessel in question. These cases are classified as Grade 3, representing confirmed operational involvement by Frontex aerial assets.

Grade 2 - Frontex Asset detected the boat and the case could be matched with Frontex Situation Centre (FSC) Aerial Surveillance operation

To further refine the attribution of Frontex involvement, a case matching process was undertaken. This method cross referenced confirmed interception incidents with separate entries from the JORA dataset that were explicitly tied to surveillance operations, particularly "FSC Aerial Surveillance 2021" and "FSC Aerial Surveillance 2022."

The aim was to identify incidents that had been reported twice within the database:

1. Once as a confirmed pushback with a recorded place of disembarkation as per the analysis described above; and
2. Once under a distinct entry describing a Frontex surveillance operation likely linked to the same event.

To implement this, all confirmed pushbacks with known disembarkation points were excluded, yielding a residual dataset. A filtered subset of surveillance-labelled operations was then compared against confirmed pushbacks based on two conditions:

- The detection date was identical; and
- The number of persons on board (PoB) matched exactly, with a difference of zero.

This conservative approach ensured each interception was paired with at most one corresponding surveillance record. Such matches, where an incident appears both as an interception and as a surveillance operation, were classified as Grade 2—indicating indirect but strongly corroborated involvement by Frontex through aerial detection and monitoring.

Grade 1: Frontex asset did not only detect but initiated the detection with their Maritime Aerial Surveillance (MAS)

The highest level of attribution, Grade 1, was assigned to incidents that satisfied the conditions of Grade 2 and where both of the following additional fields confirmed Frontex's direct role:

- DetectionInitiatedBy was listed as “MAS”; and
- DetectionByFrxAsset was marked as “Yes.”

This combination signals that the incident was initiated by Frontex’s aerial platform, not merely observed or reported post hoc. These incidents represent the strongest available evidence of Frontex’s operational role in detecting and initiating the pushback. It should be noted, however, that Grade 1 classification is not possible in the JORA 2 dataset (covering incidents from September 2021 onward), due to redactions made by Frontex in this version of the dataset. Specifically, the fields identifying who initiated the detection were no longer included, precluding the application of this final filter.

Grade	Criteria	Summary Description
Grade 4	All filtered maritime pushbacks recorded in JORA	Documented interception (baseline)
Grade 3	Grade 4 + detection by Frontex asset	Interception detected by Frontex aerial asset
Grade 2	Grade 4 + case matched with separate surveillance operation on same date and PoB	Indirect confirmation via case matching
Grade	Grade 2 + detection was initiated by Frontex	Confirmed aerial detection initiated by Frontex

Summary Table: Grading System

Results

Applying the harmonised filters yields a subset of pushback incidents that we then coded using a graded attribution scheme capturing two distinct dimensions: (i) Frontex’s operational role in the event (from awareness-only to active contribution/triggering through aerial detection or on-scene monitoring) and (ii) the strength of the supporting evidence (from minimal documentation to corroborated cases). This grading allowed us to move beyond counting incidents by distinguishing cases where Frontex merely recorded an interception from cases where available documentation indicates a plausible causal contribution to the interception and return —thereby enabling a structured assessment of Frontex’s potential degree of complicity.

	Year	Grade 4		Grade 3		Grade 2		Grade 1	
		n_boats	n_people	n_boats	n_people	n_boats	n_people	n_boats	n_people
JORA 1	2016	1	127	0	0	0	0	0	0
	2017	33	3 369	0	0	0	0	0	0
	2018	88	8 030	1	10	0	0	0	0
	2019	120	8 372	16	1 131	3	215	2	135
	2020	170	9 048	72	4 891	7	377	7	377
	2021	434	26 265	234	13 511	112	5 387	72	3 410
JORA 2	2022	253	13 997	158	8 029	91	4 151	Redacted	Redacted
	2023	11	601	5	320	1	80	Redacted	Redacted
JORA 3	2024	Redacted	Redacted	Redacted	Redacted	Redacted	Redacted	Redacted	Redacted
	2025	Redacted	Redacted	Redacted	Redacted	Redacted	Redacted	Redacted	Redacted

2. ARRIVALS

a) Establish the number of pushbacks to Libya and Tunisia recorded in Frontex JORA database

This process aimed to isolate all cases of arrivals (i.e., as defined above, cases where the disembarkation occurred in Italy or Malta) and to differentiate them by the degree of operational involvement by Frontex.

I. JORA 1

To analyse the extent of Frontex's involvement in arrivals of migrants to Italy or Malta, we applied a structured filtering and classification process to the JORA 1 dataset (2016–2021).

First, the dataset was filtered to retain only those incidents where the country of disembarkation was Italy, and country of departure was Libya or Tunisia. Given variability in place names and formatting, a comprehensive list of terms associated with Italian territory (e.g., “Italy,” “IT,” and specific port names such as “Lampedusa,” “Pozzallo,” and “Augusta”) was used to ensure exhaustive capture of all relevant incidents. This step resulted in a refined subset representing all recorded disembarkations on Italian soil, regardless of who facilitated or initiated the arrival. This subset represents Frontex’s knowledge about arrivals in Italy.

This subset of arrivals in Italy was then divided into two categories based on the operational metadata available in the JORA entries:

1. Frontex-facilitated Arrivals

Incidents were classified as monitored by Frontex when the field DetectionByFrxAsset was marked “Yes”. This designation indicates that Frontex aerial or maritime assets observed or tracked the vessel but were not necessarily the primary actors in directing or facilitating the interception or arrival. These cases reflect Frontex's surveillance presence, providing situational awareness without direct coordination.

Criteria:

- Disembarkation Country == Italy
- DetectionByFrxAsset == Yes

2. Frontex-initiated Arrivals

The highest level of involvement, Frontex-initiated arrivals, was attributed to incidents where Frontex was responsible for initiating the detection and operational response that led to the disembarkation. This was indicated by the field DetectionInitiatedBy being marked “MAS” (Maritime Aerial Surveillance), coupled with a positive entry in DetectionByFrxAsset. These entries signal that Frontex directly identified and tracked the vessel using its aerial surveillance platforms and may have played a coordinating role in ensuring its arrival in Italy.

Criteria:

- Disembarkation Country == Italy
- DetectionByFrxAsset == Yes
- °DetectionInitiatedBy == MAS

This multi-layered classification allows for a nuanced understanding of Frontex’s operational role in migrant arrivals to Italy. By distinguishing between simple monitoring, active facilitation, and direct initiation, the analysis captures the varying levels of Frontex engagement in maritime migration over the study period.

II. JORA 2

Due to the absence of an explicit field indicating disembarkation location in the second JORA dataset, arrivals in Italy or Malta were identified through a multi-step filtering process designed to isolate incidents that occurred at sea and did not result in returns to Tunisia or Libya. The method focused on maritime interceptions with a high likelihood of concluding in disembarkation on Italian or Maltese territory.

The process began with a subset of incidents involving departures from Libya or Tunisia, already identified earlier in the workflow through Filter 1. From this, incidents were retained if the interception place was recorded as either:

- "Sea"
- "Inside national territory / on shore"
- "BCP" (Border Crossing Point)

These values capture maritime or coastal activities events likely related to migrant arrivals in Italy rather than interceptions at sea pushed back to Libya.

A temporal filter was then applied: only incidents that occurred after 6 September 2021 were retained. This date reflects the cut-off corresponding to the revision of Frontex's internal data recording procedures and marks the beginning of the second dataset's timeframe.

To ensure that the remaining incidents were distinct from returns to Libya or Tunisia, all cases already included in the separately prepared "Prevention of Departure" dataset were removed by checking for overlaps in Incident ID. These represent confirmed or suspected non-arrivals and were excluded to focus the analysis solely on potential arrivals.

Additional filters were applied to eliminate events unrelated to irregular sea crossings:

- Incidents involving transport types such as ferries, cars, vans, or cargo vessels were removed.
- Any incident where the primary incident type was "Illegal fishing" was excluded, as these do not pertain to migration-related operations.

Finally, incidents where the destination country was explicitly recorded as France, Germany, Spain, or other non-Italian European states were excluded. This step ensured the final dataset focused on events not associated with redirection to other countries, thereby increasing the likelihood that they correspond to arrivals in Italy.

While the dataset does not directly confirm disembarkation in Italy or Malta, this conservative filtering strategy isolates a set of incidents with high probability of constituting arrivals on Italian shores. As the Disembarkation entry has been redacted by Frontex, we are unable to determine the exact status of each incident. However, the identification of intercepted boats in the database shows that an interception incident displays strictly the values "Prevention of departure'' in the column "interception place''. Therefore, it was assumed that all incidents departing from Libya that were not prevented from departure —and which were not intercepted by a Third Country action — arrived in Italy or Malta.

Incidents were classified as Frontex-facilitated if the field Detection by Frontex financed/deployed asset was marked as "Yes". This indicates that the detection of the vessel was carried out by a Frontex aerial or maritime surveillance asset, such

as a drone or patrol aircraft. These cases reflect direct operational monitoring by Frontex, suggesting logistical or strategic support for the interception and potential arrival.

A more active level of involvement was captured by identifying incidents where Frontex was the initiating party in the detection process. These were cases where the field Detection - Member State was set to "Frontex". This designation signifies that Frontex was solely responsible for initiating the detection, rather than acting in a supporting role to national authorities. These entries represent the strongest attribution of operational involvement to Frontex.

Results

Across JORA 1 and 2 (Sept. 2016 – Jan. 2023), the filtered JORA arrivals subset contains 7,146 arrival events in Italy or Malta, involving 361,943 people in total. These figures provide the baseline for assessing (i) how completely Frontex documents arrivals and (ii) how often Frontex assets are operationally involved (through monitoring and initiating detection), in contrast with outcomes ending in pushbacks to Libya/Tunisia.

Arrivals in Italy according to Frontex

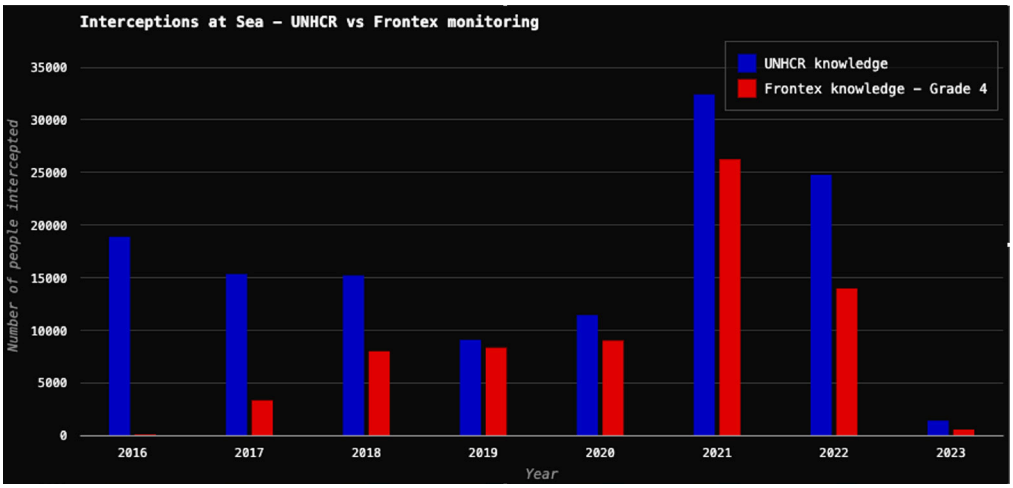
	Number of boats	Number of people
2016	447	48 787
2017	1 201	113 158
2018	431	19 327
2019	313	9 976
2020	1 095	28 610
2021	1 363	51 734
2022	2 206	86 480
2023*	90	3 871

*only until January 2023

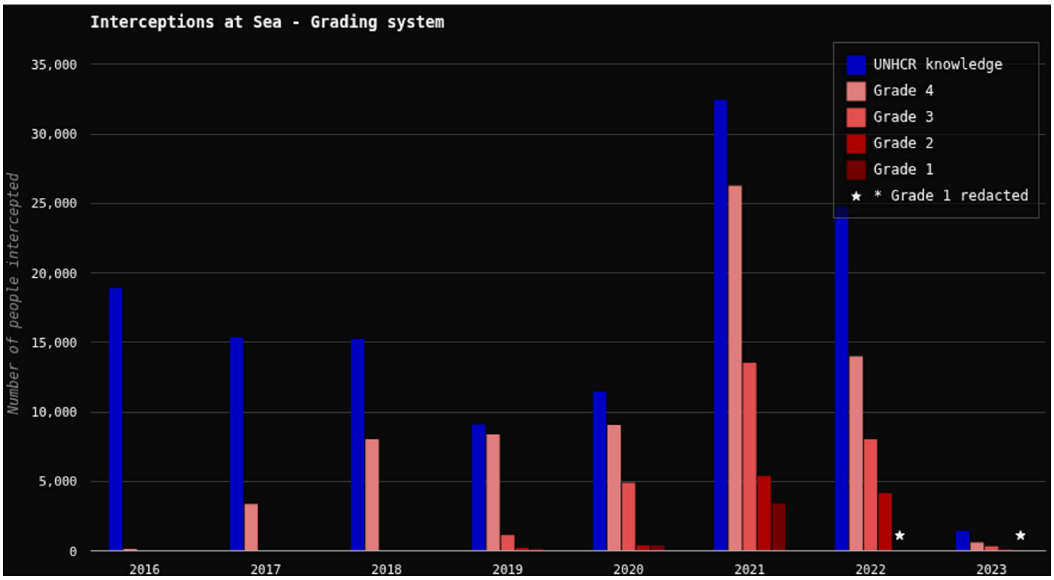
Discussion Of Results

1 Completeness of Frontex's reporting, and degree of complicity in pushbacks

This section assesses the completeness of Frontex's reporting on pushbacks by comparing, year by year, the number of intercepted people recorded in Frontex's JORA database (our Grade 4 baseline) with UNHCR interception totals. The resulting ratios provide a measure of how closely Frontex's JORA incident reporting tracks independently monitored interceptions over time.



From 2016 to 2019, there was a clear increase in the proportion of interceptions recorded by Frontex, signalling that before the establishment of a more systematic aerial surveillance mechanism the Agency's ability to monitor what was happening at sea was very low. In 2017, Frontex accounted for only **22%** of total interceptions monitored by UNHCR. By 2019, once aerial operations had started to be deployed by Frontex on a more systematic basis, this rose to **92%** – indicating a substantial improvement in Frontex's recording capabilities. However, from 2020 onwards, this proportion began to decline slightly — **79%** in 2020, **81%** in 2021, **56%** in 2022, and **42%** in January 2023. This recent decline is largely attributable to our own conservative data analysis methodologies that we decided to adopt in response to missing information and redactions within the Frontex dataset obtained, which make full visibility of events increasingly challenging.

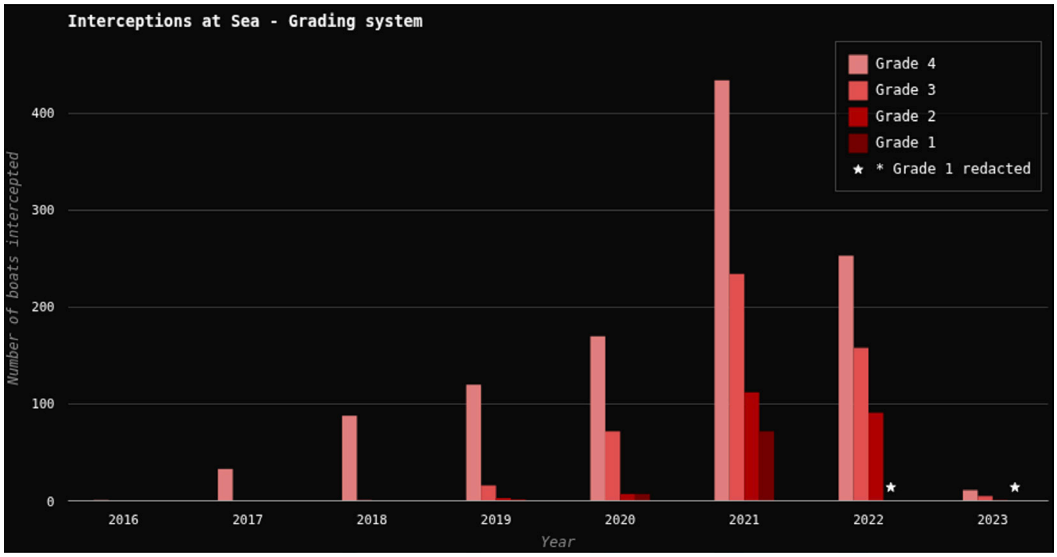


Year	UNHCR documented interception	Frontex documented interception	Frontex facilitated interception	Frontex initiated interception	% of Frontex documentation	% facilitated	% initiated
2016	18904	127	0	0	1%	0%	0%
2017	15362	3369	0	0	22%	0%	0%
2018	15235	8030	10	0	53%	0%	0%
2019	9108	8372	1131	586	92%	12%	6%
2020	11464	9048	4891	3896	79%	43%	34%
2021	32425	26265	13511	11523	81%	42%	36%
2022	24788	13997	8029	7886	56%	32%	32%
2023*	1446	601	320	320	42%	22%	22%

*only until January 2023

Importantly, the data show that Frontex has increasingly played an active role in enabling the interception of people by the Libyan Coast Guard (LYCG). In 2019, the interception of **586** people had been initiated by Frontex (6% of UNHCR-documented interceptions), growing to **3,896** in 2020 (34%), and peaking at **11,523** in 2021 (36%). The number of people intercepted during events facilitated by Frontex — i.e., where the Agency was involved in detection or tracking at some point but not direct coordination or the first sighting of vessels—also increased significantly, reaching **13,511** in 2021 (42%).

This indicates that during the 2020–2021 period, **at least 34%** of intercepted individuals were first located by a Frontex aerial asset. Even in 2022 and January 2023, despite reduced numbers overall, Frontex still initiated **32%** and **22%** of interceptions respectively, confirming an ongoing operational role.

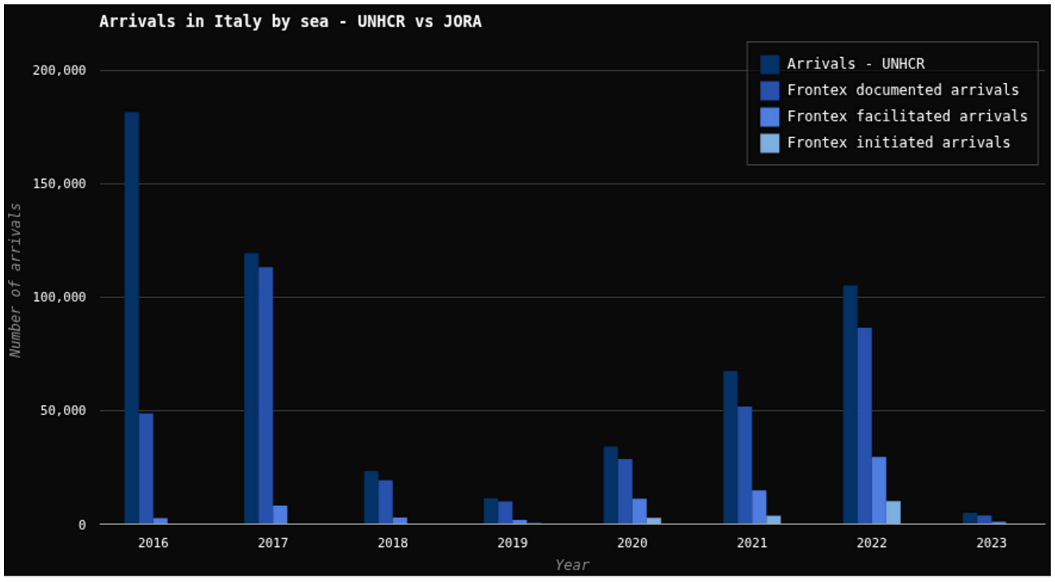


Year	Grade 4	Grade 3	Grade 2	Grade 1
2016	1	0	0	0
2017	33	0	0	0
2018	88	1	0	0
2019	120	16	3	2
2020	170	72	7	7
2021	434	234	112	72
2022	253	158	91	Redacted
2023*	11	5	1	Redacted
2024	Redacted	Redacted	Redacted	Redacted
2025	Redacted	Redacted	Redacted	Redacted

*only until January 2023

The above graph highlights the distribution of interception events according to the grading system (Grade 1 to Grade 4). Grade 4, which includes all cases of pushbacks Frontex is aware of regardless of the Agency's specific role in them, surged to **434 interception events in 2021**, the highest across the observed period. There was a subsequent drop to **253 in 2022** and **11 in 2023**, though the 2023 figure may be underrepresented due to data limitations. Grade 3 and Grade 2 cases also follow similar patterns, with a marked increase in 2020–2021, indicating a growing prevalence of Frontex's involvement in interceptions during those years.

2 Frontex involvement in pushbacks versus safe arrivals to Italy and Malta



This section examines what Frontex's aerial surveillance is used for in practice, by contrasting its operational involvement in events ending in pushbacks to Libya

with its footprint in events ending in arrivals to Italy and Malta. We compare (i) documentation rates and (ii) the share of cases where Frontex assets detect and/or initiate detection, to show whether Frontex's operational contribution is more strongly associated with pushbacks to Libya than with outcomes leading to disembarkation in a "place of safety".

In contrast to pushbacks, Frontex monitoring of arrivals in Europe (Italy and Malta) aligns closely with UNHCR records. For example, in 2017, Frontex documented **95%** of the arrivals monitored by the UNHCR. However, its operational contribution to these arrivals remains extremely limited. Across the 2016–2023 period, Frontex initiated 296 of the 7,146 arrival events identified in JORA (4.1%). These events involved 16,951 people, equivalent to 3.1% of arrivals recorded by UNHCR. In contrast, Frontex played a significant operational role in interceptions leading to returns to Libya, or pushbacks. For instance, in 2021, Frontex initiated **3,634** European disembarkations (5%), whereas it initiated **11,523** interceptions that led to returns to Libya (36%). This disparity underlines that Frontex's aerial surveillance is significantly more frequently mobilized to enable returns to Libya than to support rescues leading to safety on European shores.

Year	UNHCR_ Arrival	Frontex documented arrivals	Frontex facilitated arrivals	Frontex initiated arrivals	% Frontex documented arrivals	% Frontex facilitated arrivals	% Frontex initiated arrivals
2016	181436	48787	2634	0	27%	2%	0%
2017	119369	113158	8182	0	95%	7%	0%
2018	23370	19327	2859	0	82%	12%	0%
2019	11471	9976	1825	449	87%	16%	4%
2020	34154	28610	11186	2706	84%	32%	8%
2021	67477	51734	14912	3634	76%	22%	5%
2022	105131	86480	29539	10162	82%	28%	10%
2023	4959	3871	1083	0	78%	22%	0%

3 Opacity and concealment

The opacity in Frontex data practices increases significantly from 2023 onward. This was not a sudden development but the culmination of a progressive process of crucial data redaction. In earlier years, JORA data was relatively unredacted, therefore allowing for detailed analysis of Frontex's involvement in interceptions. From September 2021 (the start of the JORA 2 dataset), however, key variables such as the place of disembarkation were removed. This made it far more difficult to identify which incidents corresponded to pushbacks to Libya and, critically, it prevented the application of the Grade 1 category in our methodology, which requires knowledge of who initiated the detection. The most recent version of the obtained JORA, JORA 3 covering January 2023 – July 2025, goes even further: the data is fully redacted, including the removal of the dates of incidents. This is an extraordinary measure for which no convincing justification exists. To claim "protection of public security" as the reason for withholding even the date of an interception is implausible and serves no operational logic, especially considering the same information had been made available previously.

Placed in the broader context, the redaction of critical data is not a neutral bureaucratic adjustment but a direct attack on accountability. By erasing the

very variables that allow independent observers to trace responsibility, Frontex and the EU institutions it serves actively undermine human rights monitoring and obstruct the possibility of public scrutiny. This deliberate concealment directly weakens oversight at the very moment when border violence and pushbacks are intensifying in the Central Mediterranean, precisely when Frontex activities are most in need of transparency and oversight.

Conclusion

The findings of this study reveal a dual pattern in Frontex operations. On one hand, the agency has improved its aerial surveillance capacities and its ability to record maritime events. On the other hand, Frontex has increasingly deployed these capabilities to facilitate returns to Libya rather than rescues leading to safe disembarkation in Europe. This operational orientation–particularly visible during the 2020–2022 period, i.e. until the available datasets allow for meaningful analysis–raises serious humanitarian concerns, given the well-documented risks of arbitrary detention, abuse, and exploitation faced by those returned to Libya.

Frontex's contribution to rescues that result in safety in Europe remains marginal. Even at its peak, only a small fraction of arrivals in Italy and Malta were initiated by Frontex assets. By contrast, the agency has consistently played an active role in enabling interceptions that led to disembarkations in Libya. These patterns reinforce criticism from humanitarian organisations that aerial surveillance is being used as a tool of border enforcement and human rights violations rather than to safeguard human life – as is often claimed by Frontex.

Compounding this critique is Frontex's progressive redaction of crucial material found in its own operational database, JORA. While earlier datasets allowed for the meaningful attribution of responsibility for events, more recent datasets have been marked by heavy redactions targeting key data categories; first disembarkation locations, then detection initiators, and in the most recent dataset even the date of incidents. The 2023–2025 dataset now lists over 6,100 incidents involving more than 240,000 people and 240 deaths or disappearances but provides no way to establish whether Frontex was involved. To claim "protection of public security" as justification for removing such basic information is implausible. Rather, the effect–and likely the intent–is to obstruct accountability at a time when border violence in the Central Mediterranean is escalating.

Taken together, these findings show that Frontex's official narrative of "protecting" human rights during its operations is fundamentally contradicted by its practice of enabling pushbacks, while obscuring its own role in them. This combination of growing operational complicity and deliberate opacity amounts to a structural attack on accountability where Frontex continues to operate in a space of impunity – shielded from scrutiny while directly contributing to crimes against humanity.